

# The transfer function of subset-sum landscapes: rigidity of the gap series off centre

Kentaro Amauchi\*

August 9, 2026

## Abstract

For a finite set  $A$  of odd integers, the number of strict local minima of the subset-sum landscape at a target  $n$ , divided by the number of ground states, is governed at the centre  $n = \lfloor T/2 \rfloor$  by the gap series  $\Gamma(A) = \sum_i a_i 2^{-i}$ . This paper argues that this is a property of the landscape rather than of its centre, and in a sharper form: that the ratio is a universal functional of a single local observable, the logarithmic slope  $\lambda$  of the representation count at the target, through an explicit transfer function  $\Phi(\lambda) = 1 + \sum_{d \geq 1} 2^{1-N_d} e^{-\lambda^2 s_d/8} \cosh(\lambda \delta_d)$  built from the small elements of  $A$  alone; rigidity is the case  $\lambda \rightarrow 0$ . We prove the analytic ingredients — an additive bridge carrying the minor-arc estimates of the companion paper from the uniform measure to the exponential tilt, the minor-arc input in the form that bridge needs, and the threshold  $\alpha = 1$  below which the governing hypothesis fails — and assemble them into a proof skeleton for the two theorems, the remaining gap being the Edgeworth constants of a classical local limit computation. The extremal denominator of the bridge is  $q = 6$ , the same one that produces the constant  $\sqrt{3}/2$  in the companion paper. The transfer function is verified against exact computation on four profiles: the residual is 0.8–1.9% of the signal and obeys a  $c_A/k$  law whose constant is predicted, with no free parameter, by the profile alone.

## 1 Introduction

Papers 1 and 2 of this series establish that for the odd primes  $\text{lm}/\text{deg} \rightarrow \Gamma$  *at the centre* of the landscape. Whether  $\Gamma$  is an invariant of the landscape or an accident of its most symmetric point is the question this paper answers, and the answer turns out to be stronger than rigidity alone: the whole ratio is a function of one local observable. The argument and what each part rests on:

- The companion papers establish  $\text{lm}/\text{deg} \rightarrow \Gamma$  *at the centre*  $n = \lfloor T/2 \rfloor$ : paper 1 reduces it to a flatness hypothesis and paper 2 discharges that hypothesis unconditionally. The obvious question — whether  $\Gamma$  is a property of the landscape or of its centre — is what this paper answers.
- The combinatorial layer (classification, exact stratification, sandwich) was proved *for every target* in paper 1 and is formally verified; nothing there needs redoing. [\[Lean-verified\]](#)
- The minor-arc estimates of paper 2 bound  $|G_B(\theta)|$  as a function of  $\theta$  only. They do not refer to the extraction point  $m$ , which enters  $r_B(m) = \frac{1}{2\pi} \int F_B(\theta) e^{-im\theta} d\theta$  only through a phase. They therefore transfer to off-centre targets unchanged. [\[quoted from paper 2\]](#)

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\*Independent researcher.

- What is new is the middle, §4.2: an additive bridge (Lemma 4.8) carrying the minor-arc estimates from the uniform measure to the exponential tilt, the minor-arc input in the form the bridge needs (Lemma 4.10), and a tilted local limit theorem uniform over the extraction window (Proposition 4.20). The odd orders in  $\lambda$  cancel by the reflection symmetry of the landscape, which is where cosh enters.
- One consequence of quoting rather than reproving deserves saying at the outset: *the only ineffective constant in the trilogy is one and the same Siegel–Walfisz constant, and it enters each paper at exactly one point* — here, part (b) of Lemma 4.10. See §7.

## 2 Notation, and what is inherited

The definitions are those of paper 1, written so that nothing assumes  $n = T/2$ . Fix throughout an increasing sequence  $A = (a_1, \dots, a_k)$  of odd positive integers,  $T = \sigma(A)$ ,  $S_2 = \sum a_i^2$ ,  $V = S_2/4$ , and a target  $n$ . Write  $\rho = n/T$  and  $x = \frac{1}{2} - \rho$ . For  $d \geq 1$  put

$$N_d = \#\{i : a_i \leq 2d\}, \quad \sigma_d = \sum_{a_i \leq 2d} a_i, \quad s_d = \sum_{a_i \leq 2d} a_i^2, \quad \delta_d = d + \frac{\sigma_d}{2},$$

and let  $B_d = \{a_i : a_i > 2d\}$  be the layer- $d$  ensemble, of variance  $V_d = (S_2 - s_d)/4 = V - s_d/4$ .

**Definition 2.1** (local logarithmic slope).  $\lambda(n)$  is the logarithmic derivative of the representation count at the target,

$$\lambda(n) = +\frac{d}{dm} \log r_A(m) \Big|_{m=n} = \frac{1}{4} \log \frac{r_A(n+2)}{r_A(n-2)} + O(\cdot),$$

so  $\lambda > 0$  below the centre, where  $r_A$  is increasing.

*Remark 2.2* (on the sign, and why it took until paper 4 to matter). An earlier version of this definition carried the opposite sign, which made it disagree with Remark 2.3 (where  $\lambda \approx (T/2 - n)/V > 0$  below the centre) and with Remark 2.5 (where  $s/\lambda \rightarrow 1$  with  $s > 0$  below the centre). Nothing in this paper changed when the sign was corrected, and nothing could have:  $\Phi$  is an *even* function of  $\lambda$ , so the sign of  $\lambda$  is not observable anywhere in the transfer function. It becomes observable only when the cosh is split into two unequal exponentials, which is what happens under a biased measure; the discrepancy was found there and is recorded here. The convention above is the one that makes  $\lambda \approx s$ .

*Remark 2.3* (why the slope and not the offset). The Gaussian approximation  $\lambda \approx (T/2 - n)/V$  is what one writes down first, and it is wrong at the 10% level for the purposes of this paper. Substituting the measured  $\lambda$  for the Gaussian one takes the drift of the fitted coefficient across the range  $x \in [0.06, 0.30]$  from 15–18% down to 0.03–0.07%, and it dissolves a non-integer exponent that an earlier draft of this programme had reported as a finding: the exponent was an artefact of the wrong variable. This is not a cosmetic choice: every statement below is false at the stated precision if  $(T/2 - n)/V$  is substituted for  $\lambda$ .

**Definition 2.4** (the tilt).  $P_s$  is the product measure on subsets under which  $a_i$  is present with probability  $p_a = (1 + e^{sa})^{-1}$ , and  $s = s(n) > 0$  is chosen so that  $\sum_a a p_a = n$ ; thus  $P_s$  is the exponential tilt of the uniform measure whose mean sits at the target.

*Remark 2.5* (the tilt and the slope agree to first order).  $s$  and  $\lambda$  are the same quantity to leading order but not exactly:  $s$  makes the tilted *mean* equal  $n$ , whereas  $\lambda$  is the slope of  $\log r_A$  at  $n$ , and the two differ by the mean–mode gap. Measured on three profiles,

$$\frac{s(n)}{\lambda(n)} = 1 + \frac{c_A}{k} + O(k^{-2}), \quad c_A \approx 1.85 \text{ (odds)}, 2.03 \text{ (primes)}, 2.87 \text{ (squares)},$$

stable in  $x$  and clean in  $k$  over  $k = 100, \dots, 220$ . [\[experimentally confirmed; tilt\\_r088\]](#) The statements below are formulated in  $\lambda$ , and *not* in  $s$ : substituting  $s$  multiplies the residual of Table 1 by three, exactly as the  $\lambda^2$  leading behaviour of  $\Phi$  predicts. [\[experimentally confirmed; sres\\_r088\]](#)

### 3 Rigidity

**Theorem 3.1** (rigidity of the gap series). [\[proof skeleton: spec\\_t3rigid\\_r087 as amended in §4.2\]](#) Let  $P = P_k$  be the first  $k$  odd primes and let  $\rho \in (0, 1)$  be fixed. Then

$$\frac{\text{lm}_P(\lfloor \rho T \rfloor)}{\deg_P(\lfloor \rho T \rfloor)} \xrightarrow{k \rightarrow \infty} \Gamma(P),$$

uniformly for  $\rho$  in compact subsets of  $(0, 1)$ . For a general profile  $A$  the same holds under hypothesis (H) of §6.

$\Gamma(A) = \sum_i a_i 2^{-i}$  is therefore an invariant of the landscape and not of its centre. Theorem 3.1 is the case  $\lambda \rightarrow 0$  of Theorem 4.1: at fixed  $\rho$  one has  $\lambda(n) \asymp x/N \rightarrow 0$ , and  $\Phi$  is continuous at 0 with  $\Phi(0) = \Gamma$ , so

$$\Phi(\lambda) - \Gamma = O\left(\lambda^2 \sum_{d \geq 1} 2^{1-N_d} \delta_d^2\right) \rightarrow 0,$$

the sum being finite precisely by (H). The uniformity in  $\rho$  needs no new estimate: every bound in §4.2 is uniform once  $t_{\min}(\rho)$  is bounded below, and  $t_{\min}$  is continuous and positive on  $(0, 1)$ , hence bounded below on compact subsets (Remark 4.19).

[\[experimentally confirmed\]](#) The evidence the theorem is modelled on. With  $\text{ratio}(\rho, k) = \lfloor \text{lm}/\deg \rfloor / \Gamma(P_k)$ , the spread over  $\rho$  contracts monotonically:

| $k$                            | 40      | 52      | 64      | 76      | 88      | 100     |
|--------------------------------|---------|---------|---------|---------|---------|---------|
| $\max_\rho - \min_\rho$        | 0.01051 | 0.00283 | 0.00257 | 0.00162 | 0.00113 | 0.00083 |
| $\max_\rho  \text{ratio} - 1 $ | 0.00921 | 0.00276 | 0.00244 | 0.00156 | 0.00109 | 0.00081 |

over  $\rho \in [0.20, 0.50]$ , by exact integer computation. Three guards are in force throughout this paper's computations and are worth stating once: every reported point satisfies  $\deg \geq 10^4$ , so that integer granularity cannot masquerade as signal; the dynamic programme was checked against brute-force enumeration over all  $2^k$  subsets at small  $k$ ; and the exact identity  $\text{lm}(n) = \text{lm}(T - n)$  is verified at every measured target, which tests the stratification and the arithmetic together.

### 4 The transfer function

**Theorem 4.1** (transfer). [\[proof skeleton: spec\\_t3rigid\\_r087 as amended in §4.2\]](#) Let  $A = P_k$  be the first  $k$  odd primes, or a general profile satisfying (H), and let  $n = \lfloor \rho T \rfloor$  with  $\rho \in (0, 1)$  fixed. Then

$$\frac{\text{lm}_A(n)}{\deg_A(n)} = \Phi_k(\lambda(n)) + E_k, \quad \Phi_k(\lambda) = 1 + \sum_{d \geq 1} 2^{1-N_d} e^{-\lambda^2 s_d/8} \cosh(\lambda \delta_d), \quad (1)$$

with  $E_k \rightarrow 0$  as  $k \rightarrow \infty$ , uniformly for  $\rho$  in compact subsets of  $(0, 1)$ .

*Remark 4.2* (what the theorem does and does not claim about  $E_k$ ). The assertion is  $E_k \rightarrow 0$  in absolute terms, nothing sharper. The measured rate  $E_k \asymp c_A/k$  of Remark 4.7 is *not* part of the statement; it is quoted as [\[experimentally confirmed,  \$k \leq 220\$ , three profiles\]](#) and stated as an open problem in §7. Distinguishing the two matters here more than usual, because the numerical agreement is far better than what is proved.

Three features of (1) deserve to be stated before any proof.

*Remark 4.3* (the ratio sees the target only through one number).  $\Phi$  depends on  $A$  alone. The target enters only through  $\lambda(n)$ . This is a stronger statement than rigidity: rigidity says the limit does not move, (1) says how it fails to move.

*Remark 4.4* ( $\lambda = 0$  recovers paper 1). The layer weight  $e^{-\lambda^2 s_d/8}$  is 1 at  $\lambda = 0$ , so  $\Phi(0) = 1 + \sum_{d \geq 1} 2^{1-N_d}$ , which is the window identity  $W_D(A) = \Gamma(A) + (2D+1)2^{-k}$  of paper 1 — already formally verified. Numerically,  $\Phi(0) - \Gamma(A)$  is 0,  $-4.7 \cdot 10^{-14}$  and  $-2.7 \cdot 10^{-15}$  for the three sequences of Table 1.

*Remark 4.5* (each layer is a smaller ensemble). The factor  $e^{-\lambda^2 s_d/8}$  is not a fudge: it is what remains when the Gaussian normalisations of  $B_d$  and of  $A$  fail to cancel, because  $V_d = V - s_d/4$ . Expanding to second order gives the coefficient of  $\lambda^2$  as  $\delta_d^2 - s_d/4$  rather than  $\delta_d^2$ . In the notation of §2, layer  $d$  is the ensemble  $B_d$  with the small elements removed, so its variance is  $V_d = V - s_d/4$  rather than  $V$ ; the weight  $e^{-\lambda^2 s_d/8}$  is exactly the price of that difference, and Remark 4.21 derives it from the ratio of partition functions rather than fitting it. The correction is termwise smaller than the term it corrects,  $s_d/4 \leq d\sigma_d/2 < \delta_d^2$ , so every term of the series stays positive and the correction never overshoots.

## 4.1 Verification

[\[experimentally confirmed\]](#) Table 1 evaluates (1) at the measured  $\lambda(n)$  against the exactly computed  $\text{lm}/\text{deg}$ .

| $x = \frac{1}{2} - \rho$         | 0.06  | 0.08  | 0.10  | 0.15  | 0.20  | 0.25  | 0.30  |
|----------------------------------|-------|-------|-------|-------|-------|-------|-------|
| odds, $k = 220$                  | 0.84% | 0.85% | 0.85% | 0.86% | 0.88% | 0.92% | 0.97% |
| primes, $k = 220$                | 0.91% | 0.92% | 0.92% | 0.94% | 0.96% | 0.99% | 1.05% |
| $a_i \asymp i^{3/2}$ , $k = 220$ | 1.05% | 1.06% | 1.06% | 1.08% | 1.10% | 1.14% | 1.20% |
| squares, $k = 170$               | 1.66% | 1.67% | 1.68% | 1.70% | 1.74% | 1.80% | 1.89% |

Table 1: Residual  $|\text{measured} - \Phi(\lambda)|$  as a fraction of the centred signal  $\text{lm}/\text{deg} - [\text{lm}/\text{deg}]_{n=T/2}$ . Absolute residuals are  $10^{-5}$  to  $10^{-6}$ ; quoting them as significant figures of the ratio would be meaningless, since the signal itself lives in the fifth and sixth digit. The third row is the only ensemble in this paper that was *constructed to test a prediction* rather than chosen because it was natural:  $\alpha = 3/2$  was built after Remark 4.7 had been fitted on the other three, and its predicted constant  $16/7$  lies between theirs.

*Remark 4.6* (why the residual is quoted this way). Against the *uncentred* signal the same residuals read 2% to 300%, the large values occurring precisely where the signal is smallest. Neither number is wrong; the centred one is the one the theorem is about, because  $\Phi(0) = \Gamma$  is an identity and the finite- $k$  offset  $[\text{lm}/\text{deg}]_{n=T/2} - \Gamma$  is a separate quantity of size  $10^{-6}$ . We keep this remark because an earlier draft of this programme reported “agreement to five or six significant figures” for a model that was 12% wrong on the signal: the figures were real and the claim was not, because the signal itself lives in those digits.

*Remark 4.7* (the residual obeys a  $1/k$  law). Table 1 is one column of a  $k$ -scan. Writing  $R_A(k)$  for the residual in the sense of the caption, the measurement over  $k = 60, \dots, 220$  gives

$$k R_A(k) \longrightarrow \begin{cases} 1.85\% & \text{odds } (k = 60, 100, 140, 180, 220 : 2.29, 2.00, 1.91, 1.87, 1.85), \\ 2.01\% & \text{primes } (k = 100, \dots, 220 : 2.38, 2.06, 2.03, 2.01), \\ 2.82\% & \text{squares } (k = 130, \dots, 190 : 3.78, 2.85, 2.83, 2.82). \end{cases}$$

Two things follow. First, the residual is  $O(1/k)$  and therefore vanishes, which is the form Theorem 4.1 asserts; the table's 1% is a finite- $k$  number and not a floor. Second, the squares row is the loosest of the three because  $c_A$  is larger, not because the decay is different: squares obey the same law with a constant 1.52 times that of the odd numbers. The constants  $c_A$  coincide numerically with those of Remark 2.5 — the same mean–mode gap seen twice — but substituting  $s$  for  $\lambda$  does not remove the residual, so the coincidence is a shared order of magnitude and not a shared cause. [\[experimentally confirmed; tilt\\_r088, sres\\_r088\]](#)

What they share is a driver. The saddle-point representation gives the exact identity

$$s(n) - |\lambda(n)| = \frac{K'''(s)}{2K''(s)^2}, \quad K'' = \sum a^2 p_a (1 - p_a), \quad K''' = \sum a^3 p_a (1 - p_a)(1 - 2p_a),$$

tested pointwise to a relative error  $0.54/k$  [\[experimentally confirmed; saddle\\_r092\]](#), and for small tilt  $K'''/(2K''^2) \approx sS_4/S_2^2$ , so that  $c'_A = k(s/|\lambda| - 1) \rightarrow kS_4/S_2^2$ . For  $a_i \asymp i^\alpha$  this is  $(2\alpha + 1)^2/(4\alpha + 1)$ . The measured  $c_A$  of this remark agrees with  $c'_A$  to within 1% on four profiles:

|                      | $\alpha$ | $kS_4/S_2^2$ | $c'_A$ | $c_A$  | $c_A/c'_A$ |
|----------------------|----------|--------------|--------|--------|------------|
| odd numbers          | 1        | 1.8000       | 1.8525 | 1.8674 | 1.008      |
| $a_i \asymp i^{3/2}$ | 3/2      | 2.2794       | 2.3494 | 2.3318 | 0.993      |
| squares              | 2        | 2.7714       | 2.8615 | 2.8377 | 0.992      |
| primes               | —        | 1.9731       | 2.0251 | 2.0253 | 1.000      |

at  $k = 220$ . The two halves of this have different statuses and the difference matters:

- $c'_A \rightarrow (2\alpha + 1)^2/(4\alpha + 1)$  is *derived* — the identity above is exact, its sign is forced by  $\Lambda''' < 0$  for  $\rho < \frac{1}{2}$ , and the small-tilt step is a Taylor expansion — and [\[experimentally confirmed on five profiles, one of them out of sample\]](#).
- $c_A = c'_A(1 + o(1))$  is [\[experimentally confirmed: four profiles, ratios 0.992–1.008,  \$k = 220\$ \]](#). The exact asymptotic equality, and the mechanism behind it, remain a **conjecture**. What is established is a shared driver,  $S_4/S_2^2$ , and *not* a causal link: substituting  $s$  for  $\lambda$  makes the residual three times worse, not smaller.

Carried to  $k = 300$ , the identification tightens rather than drifting, and it does so at a measurable rate. The ratio  $c_A/c'_A$  reads

| $k$                  | 120    | 180    | 220    | 260    | 300    |
|----------------------|--------|--------|--------|--------|--------|
| odd numbers          | 1.0589 | 1.0244 | 1.0151 | 1.0098 | 1.0064 |
| primes               | 1.0318 | 1.0084 | 1.0039 | 1.0014 | 1.0001 |
| $a_i \asymp i^{3/2}$ | 1.0113 | 1.0015 | 0.9995 | 0.9984 | —      |
| squares              | 1.0782 | 1.0039 | 0.9985 | —      | —      |

so  $|c_A/c'_A - 1|$  falls by a factor 9.2 across a factor 2.5 in  $k$  for the odd numbers, an exponent of 2.4 — the gap between the two constants closes *faster* than the  $1/k$  law they both sit on. [\[experimentally confirmed, four profiles,  \$k \leq 300\$ ; e8\\_r103\]](#) This is a sharper statement of the same conjecture, not a proof of it: a rate of empirical convergence is evidence about a limit, and the mechanism is still missing.

## 4.2 The bridge to paper 2

Both theorems reduce to the same five parts. **P1** is the exact combinatorial stratification, proved for every target in paper 1 [\[Lean-verified\]](#); **P3** is a local limit theorem for the tilted ensemble of Definition 2.4; **P4** is the assembly, in which the odd orders in  $\lambda$  cancel by the reflection symmetry  $\text{Im}(n) = \text{Im}(T-n)$  [\[Lean-verified\]](#); **P5** is the error budget, which converges exactly under (H). Only **P2** is new: the minor-arc estimates of paper 2 are proved for the *uniform* measure, and P3 needs them for the tilted one.

Write  $\tilde{G}_B(\theta) = \prod_{a \in B} (1 - p_a + p_a e(a\theta))$  for the tilted generating function and  $S_B(\theta) = \sum_{a \in B} e(a\theta)$  for the linear exponential sum.

**Lemma 4.8** (the bridge). *Put  $t_a = 4p_a(1 - p_a)$  and  $t_{\min} = \min_{a \in B} t_a$ . For every  $\theta$ ,*

$$|\tilde{G}_B(\theta)|^2 = \prod_{a \in B} (1 - t_a \sin^2(\pi a\theta)) \leq \exp\left(-t_{\min} \sum_{a \in B} \sin^2(\pi a\theta)\right) = \exp\left(-\frac{t_{\min}}{2}(b - \text{Re } S_B(\theta))\right).$$

*Consequently, if  $\text{Re } S_B(\theta) \leq \eta b$  then  $|\tilde{G}_B(\theta)| \leq \exp(-t_{\min}(1 - \eta)b/4)$ .*

*Proof.* The identity  $|1 - p + pe(\phi)|^2 = 1 - 4p(1 - p) \sin^2(\pi \phi)$  is a computation;  $t_a \geq t_{\min}$  and  $1 - u \leq e^{-u}$  give the middle step; and  $\sum_a \sin^2(\pi a\theta) = \frac{1}{2}(b - \text{Re } S_B(\theta))$  is the double-angle formula summed.  $\square$

*Remark 4.9* (why the bound is additive, and a bound that is not available). It is tempting to transfer multiplicatively, as  $|\tilde{G}_B(\theta)| \leq |G_B(\theta)|^\kappa$  with  $G_B$  the untilted product, since that would carry every exponent of paper 2 across at once. **No such  $\kappa > 0$  exists.** At any  $\theta$  with  $a\theta \in \frac{1}{2} + \mathbb{Z}$  for some  $a \in B$  the untilted factor  $\cos(\pi a\theta)$  vanishes, so the right-hand side is 0, while the tilted factor equals  $\sqrt{1 - t_a} > 0$ . Equivalently  $\log(1 - ty)/\log(1 - y) \rightarrow 0$  as  $y \rightarrow 1$ . The underlying inequality  $1 - ty \leq (1 - y)^t$  is false for every  $t \in (0, 1)$  —  $s \mapsto (1 - y)^s$  is convex and therefore lies *below* its chord  $1 - sy$ , with deficit exactly  $-(1 - t)$  at  $y = 1$ . Lemma 4.8 avoids the issue by never dividing by a quantity that vanishes. [\[refuted numerically on a 40 575-point grid and structurally; kappa\\_r088\]](#)

**Lemma 4.10** (the minor-arc input). [\[\(a\) proved, Lemma 4.11; \(b\) proved, Lemma 4.12; \(c\) quoted from paper 2\]](#) Let  $A = P_k$  and  $N = a_k$ . Then

$$\text{Re } S_A(\theta) \leq \left(\frac{1}{2} + o(1)\right)b \quad \text{uniformly for } \frac{1}{2N} \leq \theta \leq 1 - \frac{1}{2N}.$$

(a)  $q \in \{1, 2\}$ . [Lemma 4.11](#) below.

(b)  $3 \leq q \leq (\log N)^{A_0}$ . [Lemma 4.12](#) below.

(c)  $q > (\log N)^{A_0}$ . This is the case  $n = 1$  of Step 2 of `prop:deepminor` in paper 2, which gives  $|S_A(\theta)| \ll N(\log N)^{3-A_0/2} + N^{4/5}(\log N)^3$ , i.e.  $\text{Re } S_A/b \ll (\log N)^{4-A_0/2} = o(1)$  for  $A_0 > 8$ . [\[quoted from paper 2, verbatim\]](#)

**Lemma 4.11** (the two degenerate denominators). Let  $q \in \{1, 2\}$ ,  $\theta = a/q + \beta$  with  $\|\beta\| \geq 1/(2N)$ . Then  $\text{Re } S_A(\theta) \leq \frac{1}{2}b$  for all large  $N$ .

*Proof.* Write  $S_A(\theta) = \sum_{p \leq N} e(pa/q)e(p\beta)$ . For  $q = 1$  the first factor is 1; for  $q = 2$  it is  $(-1)^p = -1$  on every odd prime, so  $S_A = -\sum_{p \leq N} e(p\beta)$  and it suffices in both cases to bound  $|\sum_{p \leq N} e(p\beta)|$ . Partial summation against  $\pi(t) = \text{li}(t) + O(te^{-c\sqrt{\log t}})$  gives

$$\sum_{p \leq N} e(p\beta) = \int_2^N e(t\beta) \frac{dt}{\log t} + O(Ne^{-c\sqrt{\log N}}(1 + |\beta|N)),$$

and integrating by parts once more,  $|\int_2^N e(t\beta) dt / \log t| \ll 1/(|\beta| \log N)$ . With  $\|\beta\| \geq 1/(2N)$  this is  $\ll N/\log N \asymp b$  with an absolute implied constant, and the constant is  $\leq \frac{1}{2}$  once  $N\|\beta\|$  exceeds an absolute threshold; below that threshold, i.e. for  $\|\beta\| \asymp 1/N$ , the integral is  $(1 + o(1))b \cdot \hat{\chi}(\beta N)$  with  $\hat{\chi}$  the Fourier transform of the indicator of  $[0, 1]$ , whose real part is  $\leq \frac{1}{2}$  off the principal arc. For  $q = 2$  the sign is reversed, so  $\text{Re } S_A \leq 0$  near  $\theta = \frac{1}{2}$  — the helpful direction. No Siegel–Walfisz is used.  $\square$

**Lemma 4.12** (the small- $q$  ring). *Let  $3 \leq q \leq (\log N)^{A_0}$ ,  $(a, q) = 1$ , and  $\theta = a/q + \beta$  with  $|\beta| \leq 1/(q\tau)$ ,  $\tau = N(\log N)^{-A_0}$ . Then*

$$S_A(\theta) = \frac{\mu(q)}{\varphi(q)} W(\beta) + O(Ne^{-c\sqrt{\log N}}), \quad W(\beta) = \int_2^N e(t\beta) \frac{dt}{\log t},$$

and  $|W(\beta)| \leq \text{li}(N) = (1 + o(1))b$ . Consequently  $\text{Re } S_A(\theta) \leq (\frac{1}{2} + o(1))b$ .

*Proof.* Sort the primes  $p \leq N$  by their residue class. The  $\omega(q) = O(\log q)$  primes dividing  $q$  contribute  $O(\log q) = o(b)$ , so

$$S_A(\theta) = \sum_{\substack{r \bmod q \\ (r, q) = 1}} e\left(\frac{ra}{q}\right) \sum_{\substack{p \leq N \\ p \equiv r \pmod{q}}} e(p\beta) + O(\log q).$$

By Siegel–Walfisz,  $\pi(t; q, r) = \text{li}(t)/\varphi(q) + O(te^{-c\sqrt{\log t}})$  uniformly for  $q \leq (\log N)^{A_0}$ ; partial summation against  $e(t\beta)$  turns this into

$$\sum_{\substack{p \leq N \\ p \equiv r \pmod{q}}} e(p\beta) = \frac{W(\beta)}{\varphi(q)} + O(Ne^{-c\sqrt{\log N}}(1 + |\beta|N)),$$

and  $|\beta|N \leq (\log N)^{A_0}/q$ , so the error is  $O(Ne^{-c\sqrt{\log N}}(\log N)^{A_0}) = O(Ne^{-c'\sqrt{\log N}}) = o(b)$ : a fixed power of  $\log N$  cannot compete with  $e^{-c\sqrt{\log N}}$ . Summing over  $r$  and using the Ramanujan sum  $\sum_{(r, q) = 1} e(ra/q) = c_q(a) = \mu(q)$  for  $(a, q) = 1$  gives the display. Finally  $\mu(q)/\varphi(q)$  is real and  $\varphi(q) \geq 2$  for  $q \geq 3$ , so  $\text{Re } S_A(\theta) \leq (|\mu(q)|/\varphi(q))b(1 + o(1)) \leq (\frac{1}{2} + o(1))b$ ; Lemma 4.17 says where the  $\frac{1}{2}$  is attained.  $\square$

*Remark 4.13* (where the ineffectivity is). The Siegel–Walfisz step in Lemma 4.12 is the only inexplicit estimate in §4.2 — Lemmas 4.8, 4.11, 4.17 and the Vaughan quotation in Lemma 4.10(c) are all effective — and it is the same constant that paper 2 isolates as  $C_1$ ; see Remark 7.1.

*Remark 4.14* (what would make  $C_1$  effective). Two escapes are worth recording, because  $C_1$  is the only inexplicit quantity in the trilogy. Under GRH, the Siegel–Walfisz input of Lemma 4.12 may be replaced by  $\pi(t; q, r) = \text{li}(t)/\varphi(q) + O(t^{1/2} \log t)$  with an absolute implied constant, uniformly for  $q \leq t^{1/2}$ . The range Lemma 4.12 actually uses,  $q \leq (\log N)^{A_0}$ , is far inside that, so on GRH the constant becomes effective and every constant in the trilogy is explicit. Unconditionally, the ineffectivity is confined to the possible existence of a Siegel zero



for a single modulus  $q \leq (\log N)^{A_0}$ : for every modulus with an explicit zero-free region the estimate is already effective. The inexplicit constant is therefore not a feature of the argument but a hostage to one hypothetical zero, and the range it is held over is only  $(\log N)^{A_0}$ , not  $N$ . Neither escape is used below; the theorems are stated unconditionally.

*Remark 4.15* (the middle range, and why no constant enters). Inside  $|\theta| < 1/(2N)$  Lemma 4.10 says nothing, and it does not have to: there  $|a\theta| \leq \frac{1}{2}$  for every  $a \in A$ , so  $\sin^2(\pi a\theta) \geq 4a^2\theta^2$  and Lemma 4.8 gives the quadratic bound  $|\tilde{G}_B(\theta)| \leq \exp(-4t_{\min}V_d\theta^2)$  directly. Cutting at the LCLT-natural radius  $\theta_0 = (\log k)/\sqrt{K_d''}$  of §4.3, this is  $\leq \exp(-t_{\min}(\log k)^2)$  on the whole of  $\theta_0 \leq |\theta| \leq 1/(2N)$  — superpolynomially below the density scale  $(K_d'')^{-1/2}$ , with no free constant anywhere. The crossover at  $\theta = 1/(2N)$  survives as the boundary against Lemma 4.10, forced by  $|a\theta| \leq \frac{1}{2}$  rather than chosen; measured, the quadratic bound has already reached  $e^{-0.01b}$  by  $\theta N = 0.18\text{--}0.42$  across three profiles. [\[experimentally confirmed; gap\\_r090\]](#)

*Remark 4.16* (what part (c) does *not* need). Paper 2 must control every harmonic  $n\theta$  with  $n \leq N_0$ , and pays for it with a covering lemma and the parameter  $Q_2 = (\log N)^{A_0+A'_0}$ . Here only  $n = 1$  occurs, so the dichotomy in Lemma 4.10 is on the single denominator  $q_1(\theta)$  and no covering lemma is needed. The quotation in (c) is therefore valid on a strictly larger set of  $\theta$  than paper 2's type (ii).

**Lemma 4.17** (the extremiser of the additive estimate).  $\max_{q \geq 3} \frac{\mu(q)}{\varphi(q)} = \frac{1}{2}$ , and the maximum is attained at  $q = 6$  and at no other  $q$ .

*Proof.*  $\varphi(q) = 2$  holds exactly for  $q \in \{3, 4, 6\}$ , where  $\mu$  takes the values  $-1, 0, +1$ ; so among these the maximum is  $\mu(6)/\varphi(6) = 1/2$ . Every other  $q \geq 3$  has  $\varphi(q) \geq 4$  and hence  $\mu(q)/\varphi(q) \leq 1/4 < 1/2$ .  $\square$

*Remark 4.18* (the same  $q = 6$ , twice). By Lemma 4.10(b) the worst point of the additive estimate is therefore  $\theta = 1/6$ , with  $\operatorname{Re} S_A/b \rightarrow \frac{1}{2} = \cos 60^\circ$ . Paper 2's multiplicative estimate has its worst point at the same  $q$ :  $\sup_q M(q) = M(6) = \sqrt{3}/2 = \cos 30^\circ$ . **Same extremiser, same cause, two functionals** — both say that the odd primes lie in  $\pm 1 \pmod{6}$  and nowhere else, read once through  $\prod |\cos|$  and once through  $\sum \cos$ . Measured on the odd primes below  $10^7$ ,  $\operatorname{Re} S_A(1/6)/b = 0.499998$  against the predicted  $1/2$ , and  $q = 6$  is the maximiser over  $2 \leq q \leq 16$  at each of  $N = 10^5, 10^6, 10^7$ . [\[experimentally confirmed; eta\\_r090\]](#)

*Remark 4.19* (the honest form of the tilt constant).  $t_{\min} = t_{\min}(\rho)$  is positive for every fixed  $\rho \in (0, 1)$  and bounded below on compact subsets, which is all Lemma 4.8 needs. It is *not* governed by the approximation  $sN \approx 6x$ : that relation is the small- $x$  asymptote only, and the measured ratio  $sN/6x$  is  $1.02\text{--}1.13$  at  $x = 0.10$  but  $1.22\text{--}1.39$  at  $x = 0.30$ . Measured values of  $t_{\min}$ :  $0.91, 0.89, 0.91$  at  $x = 0.10$  and  $0.36, 0.28, 0.34$  at  $x = 0.30$  for the odd numbers, squares and primes. [\[experimentally confirmed; tilt\\_r088, kappa\\_r088\]](#)

### 4.3 The tilted local limit theorem, and the assembly

Write  $K_d'' = \sum_{a \in B_d} a^2 p_a (1 - p_a)$  and  $\mu_d(s) = \sum_{a \in B_d} a p_a$  for the variance and mean of  $B_d$  under  $P_s$ , and  $K_d^{(j)}$  for the higher cumulants.

**Proposition 4.20** (tilted LCLT, uniform over the window). [\[R2 and R3 proved \(they quote Lemmas 4.8 and 4.10\); R1 is the classical local limit computation with  \$p\_a\$ -weights, written here to the level of its cumulant budget\]](#) For every layer  $d \leq D(k)$  and every  $m$  with  $|m - \mu_d(s)| \leq W(k)$ ,

$$P_s[\sum_{B_d} = m] = \frac{1}{\sqrt{2\pi K_d''}} \exp\left(-\frac{(m - \mu_d)^2}{2K_d''}\right) (1 + O(\varepsilon)), \quad \varepsilon = \frac{|K_d'''|}{(K_d'')^{3/2}} + \frac{K_d^{(4)}}{(K_d'')^2} + \frac{W(k)|K_d'''|}{(K_d'')^2},$$



uniformly in  $d$  and  $m$ .

*Proof sketch.* Invert  $\tilde{G}_{B_d}$  over  $[-\pi, \pi]$  and split at  $\theta_0 = (\log k)/\sqrt{K_d''}$  and at  $1/(2N)$ .

*R1*,  $|\theta| \leq \theta_0$ . Expand  $\log \tilde{G}_{B_d}(\theta) = i\mu_d\theta - \frac{1}{2}K_d''\theta^2 - \frac{i}{6}K_d'''\theta^3 + \frac{1}{24}K_d^{(4)}\theta^4 + O(K_d^{(5)}\theta^5)$ . Since  $|K_d^{(j)}| \leq \sum_a a^j$ , on  $|\theta| \leq \theta_0$  every term beyond the quadratic is  $o(1)$  and the standard Edgeworth argument gives the display with the stated  $\varepsilon$ ; the third term contributes the  $W|K_d'''|/(K_d'')^2$  piece over a window of width  $W$ .

*R2*,  $\theta_0 \leq |\theta| \leq 1/(2N)$ . Remark 4.15: the contribution is  $\leq \exp(-t_{\min}(\log k)^2)$ , super-polynomially below  $(K_d'')^{-1/2}$ .

*R3*,  $|\theta| \geq 1/(2N)$ . Lemmas 4.8 and 4.10:  $|\tilde{G}_{B_d}| \leq \exp(-t_{\min}(1-\eta)b/4)$ , exponentially small, against measure  $O(1)$ .

Uniformity in  $d$ :  $B_d$  differs from  $A$  in  $N_d \leq N_{D(k)} = (\log k)^{O(1)}$  elements of size  $\leq 2D(k)$ , so every cumulant shifts by a relative  $O(s_D/S_2) \rightarrow 0$  and one  $\varepsilon$  serves all layers. The lattice is of step 1 and the parity weighting is quoted from paper 2.  $\square$

*Remark 4.21* (the  $\mu_d$ -shift, and why the layer term is what it is). This is the step that produces  $\Phi$ , so it is worth doing once in full. Since  $\sum_{a \in A} a p_a = n$  and  $p_a = \frac{1}{2} - \frac{sa}{4} + O((sa)^3)$  for the small elements,

$$\mu_d(s) = n - \sum_{a \leq 2d} a p_a = n - \frac{\sigma_d}{2} + w + \dots, \quad w := \frac{s s_d}{4},$$

so the two evaluation points of the stratification sit at

$$(n+d) - \mu_d = +\delta_d - w, \quad (n-d-\sigma_d) - \mu_d = -\delta_d - w. \quad (2)$$

Feeding (2) into Proposition 4.20, undoing the tilt through  $r_{B_d}(m) = P_s[\sum_{B_d} = m] Z_{B_d}(s) e^{sm}$ , and using  $sd + s\sigma_d/2 = s\delta_d$  to pair the two points:

$$r_{B_d}(n+d) + r_{B_d}(n-d-\sigma_d) = 2 Z_{B_d} e^{sn-s\sigma_d/2} \frac{e^{-(\delta_d^2+w^2)/2K_d''}}{\sqrt{2\pi K_d''}} \cosh\left(\delta_d\left(s + \frac{w}{K_d''}\right)\right).$$

Dividing by  $r_A(n) \sim Z_A e^{sn}/\sqrt{2\pi K''}$  and using  $Z_{B_d} e^{-s\sigma_d/2}/Z_A = 2^{-N_d} \prod_{a \leq 2d} \cosh(sa/2)^{-1} = 2^{-N_d} e^{-s^2 s_d/8} (1 + \dots)$ ,

$$\frac{r_{B_d}(n+d) + r_{B_d}(n-d-\sigma_d)}{r_A(n)} = 2^{1-N_d} e^{-\lambda^2 s_d/8} \cosh(\lambda \delta_d) \cdot e^{-\delta_d^2/2K_d''} (1 + O(\varepsilon)), \quad (3)$$

with  $\lambda = s + w/K_d'' = s(1 + s_d/(4K_d''))$  — which is exactly the per-layer slope drift of  $1 + O(s_d/S_2)$  measured earlier. **The first three factors of (3) are the general term of  $\Phi$  in (1)**; the odd orders in  $\lambda$  cancel because the two points are  $\pm\delta_d$  about the same centre, which is the reflection symmetry  $\text{lm}(n) = \text{lm}(T-n)$  in analytic dress. [\[Lean-verified \(the symmetry\)\]](#)

*Remark 4.22* (the one factor  $\Phi$  omits). (3) carries a fourth factor,  $c_d := e^{-\delta_d^2/2K_d''}$ , that  $\Phi$  does not. It is  $\lambda$ -independent and equal to  $1 - O(\delta_d^2/V)$ , so it belongs to  $E_k$  and not to the transfer function. Measured at  $k = 180$ : the ratio of the left side of (3) to  $\Phi$ 's term is  $c_d$  to six decimal places at  $d = 1$  and  $d = 3$  (1.000000 and 1.000003 for the odd numbers, 1.000000 and 1.000000 for the primes), and departs from 1 only at  $d = D(k)$ , where the layer weight is  $2^{-29} \approx 2 \cdot 10^{-9}$ . Carrying  $c_d$  inside  $\Phi$  would move Table 1 from 0.842% to 0.814% (odd numbers,  $k = 220$ ,  $x = 0.06$ ), from 0.914% to 0.902% (primes) and from 1.052% to 1.048% ( $\alpha = 3/2$ ): the factor is real, it moves the residual the right way, and it accounts for at

most 3% of it. We therefore keep  $\Phi$  as stated and count  $c_d$  in  $E_k$ . [\[experimentally confirmed; p3\\_r096, phig\\_r096\]](#)

That choice was made at one value of  $k$ , and the level of the correction is the wrong thing to decide it on; the trend is the right thing. Since  $1 - c_d \asymp \delta_d^2/2V_d$  with  $V \asymp k^{2\alpha+1}$ , and the series is dominated by the small  $d$  where  $\delta_d = O(1)$ , the  $c_d$  correction is  $O(k^{-(2\alpha+1)})$  while the residual is  $O(1/k)$ : the *share* of the residual that  $c_d$  carries should fall like  $k^{-2\alpha}$ . It does. Writing that share as  $(R_\Phi - R_{\Phi_c})/R_\Phi$  and fitting successive  $k$ :

$$1.98 \text{ (odd numbers, } \alpha = 1), \quad 2.35 \text{ (primes),} \quad 2.99 \text{ (} \alpha = \frac{3}{2}),$$

against the predicted  $2\alpha$ , i.e. 2.00,  $\approx 2.4$  for  $a_i \asymp i \log i$ , and 3.00; for the squares the prediction is  $k^{-4}$  and the correction is already at  $1 - c_1 = 8.9 \cdot 10^{-10}$ , so its share is numerical noise and changes sign — the prediction taken to its limit rather than a counterexample. Concretely the share for the odd numbers runs 10.74%, 4.93%, 3.33%, 2.40%, 1.81% at  $k = 120, 180, 220, 260, 300$ . [\[experimentally confirmed, four profiles,  \$k \leq 300\$ ; e8\\_r103\]](#) Keeping  $\Phi$  as printed is therefore not a convenience at the  $k$  we happened to measure:  $c_d$ 's contribution is a vanishing fraction of a vanishing residual, so  $\Phi$  is the right transfer function for every larger  $k$  and  $c_d$  belongs to  $E_k$  permanently. The fork is closed.

[\[experimentally confirmed; p3\\_r096\]](#) The verification behind Proposition 4.20 and (2), at  $k = 180$ , two profiles,  $x = 0.10$  and  $x = 0.30$ , layers  $d = 1, 3, D(k)$ : the  $\mu_d$  expression is exact to a relative  $6 \cdot 10^{-5}$  or better; the two offsets of (2) to  $7.6 \cdot 10^{-4}$  of  $\delta_d$  or better; the R1 formula against the exact  $r_{B_d}$  over the window to  $2.7 \cdot 10^{-3}$ – $3.7 \cdot 10^{-2}$ , in every case between 4% and 14% of the predicted  $\varepsilon$ ; the parity oscillation of  $r_{B_d}$  over the window at most  $3.7 \cdot 10^{-3}$ ; and  $\varepsilon$  itself grows by at most 44% from  $d = 1$  to  $d = D(k)$ , so a single  $\varepsilon$  does serve all layers. With Lemmas 4.8 and 4.10 the tilted generating function decays like  $\exp(-t_{\min}(1 - \eta)b/4)$  off the principal arc — exponentially in  $b$ , which is what P5 quotes. The tilted analogue of the  $M(q)$  spectrum is lost, and is not needed: the rate enters Theorem 4.1 only through  $E_k = o(1)$ .

*Remark 4.23* (what closing the skeleton would take, and where to look). What Proposition 4.20 leaves open is not a mechanism but a list of explicit Edgeworth constants in region R1, and it is worth saying that there are two ways to get them. Deriving them by hand from the characteristic function is possible and unpleasant. Quoting them may not be:  $\sigma(S) = \sum a_i X_i$  is a *weighted* sum of independent Bernoulli variables, and that is precisely the class for which the Bernoulli-part-extraction method yields local limit theorems with explicit universal constants for non-identically distributed lattice summands. Two features of Proposition 4.20 are absent from the standard statements and would have to be supplied: the summands are tilted, and the conclusion must be uniform over the extraction window and across the layers  $B_d$ . The first is harmless, since the tilt only changes  $p_a$ ; the second is what check (D) of §4.3 measures, and it found the layer spread to be at most 44%. **[TODO this remark records a route, not a result. The leads are the Bernoulli-part-extraction literature (Giuliano–Weber and the subsequent work on general weighted Bernoulli sums); neither has been read against Proposition 4.20, and until one has, nothing here may be cited as a proof.]**

## 5 Universality of the rate function

[\[experimentally confirmed,  \$k = 16..56\$ , three reference sequences\]](#) With  $\hat{I}_A(\rho) = -k^{-1} \log(\deg_A(\lfloor \rho T \rfloor)/2^k)$  and  $D_k = \hat{I}_P - \hat{I}_{\text{ref}}$ , the difference tends to 0 for each of three reference sequences: the odd numbers, the arithmetic progression  $6n + 5$ , and a random odd sequence of the same range. The sign is a property of the reference, not a universal fact — it is positive against the odd numbers, negative against the random sequence, and changes sign near  $k = 48$  against the progression.

*Remark 5.1* (the primes sit near the random sequence). At  $k = 56$  the distance to a random odd sequence is 0.0016 against 0.0146 to the arithmetic progression — an order of magnitude. The reading we propose is that the rate function does not see the arithmetic of  $A$  beyond its counting function: the primes are close to a random sequence of the same density because, for this observable, that is what they are. It is the same statement the transfer function makes locally, since  $\Phi$  depends on  $A$  only through  $(N_d, \sigma_d, s_d)$  — the small elements, counted. This is the most direct evidence we have for the reading that the main term is universal and the arithmetic of the sequence enters only through the correction.

## 6 A profile for which the hypothesis fails

The hypothesis Theorem 3.1 needs is the convergence, with room, of the series in (1) — qualitatively, that  $N_d \rightarrow \infty$  fast enough. Precisely:

$$(H) \quad \sup_k \sum_{d \geq 1} 2^{-N_A(d)} \delta_d^2 < \infty, \quad \text{and} \quad W(k) := \max\{\delta_d : 2^{-N_d} \delta_d^2 \geq k^{-1}\} = N^{o(1)}.$$

The first clause makes  $\Phi$  converge and makes  $\Phi(\lambda) - \Gamma = O(\lambda^2)$ ; the second is what the error budget needs, since the extraction window must be narrow enough for Proposition 4.20 to be uniform across it. For the odd primes  $N_A(d) = \pi(2d) - 1 \gg d/\log d$ , so both clauses hold, with  $W(k) = O((\log k)^{2+\epsilon})$ .

[\[experimentally confirmed\]](#) It is not vacuous. For the odd-ified cubes  $a_i = 2\lfloor i^3/2 \rfloor + 1$ :

| profile                                                 | odds | primes | squares | cubes          |
|---------------------------------------------------------|------|--------|---------|----------------|
| $N_d$ at $d = 10$ (any $k$ )                            | 10   | 7      | 4       | <b>2</b>       |
| $Q(0) = \Gamma^{-1} \sum 2^{-N_d} (\delta_d^2 - s_d/4)$ | 20.3 | 50.4   | 916     | <b>381 904</b> |

Only  $\{1, 9\}$  ever lie below 20, at any  $k$ , so  $2^{-N_d}$  never damps the growing  $\delta_d^2$  and the series turns over only near  $d \approx 10^4$ . At  $k = 70$  the finite-size offset is still  $1.75 \cdot 10^{-2}$  while the signal is  $1.7 \cdot 10^{-11}$ : nine orders of magnitude, so the profile is not merely slow but undecidable at any accessible size. The cubes are therefore not a defect of the theory but the reason (H) has to be there: without it, nothing in §4.2 or §4.3 fails, and yet the conclusion is out of reach.

### 6.1 Two different kinds of failure

The cubes fail *effectively* and not asymptotically, and the distinction is worth keeping. Computed directly,  $\sum_d 2^{-N_d} \delta_d^2$  for the odd-ified cubes settles at  $1.112 \cdot 10^7$  and does not grow with  $k$  (values at  $k = 60, 100, 140, 180$ :  $1.112 \cdot 10^7$  from  $k \approx 140$  on, against 69.5 for the odd numbers, 266.1 for  $\alpha = 3/2$  and 6473 for the squares — each of them  $k$ -independent as well). [\[experimentally confirmed; alpha\\_r092\]](#) So (H) is not what the cubes violate; what they violate is any hope of seeing the asymptotics, the constant being four orders of magnitude larger than the odd numbers’.

A profile that fails (H) outright is available, and on the other side:

**Proposition 6.1** (the  $\alpha = 1$  threshold). *Let  $A = A_k$  consist of  $k$  distinct odd integers with  $a_i \asymp c i^\alpha$ .*

(i) *If  $\alpha < 1$  then (H) fails:  $\sum_{d \geq 1} 2^{-N_d} \delta_d^2 \rightarrow \infty$ .*

(ii) *If  $\alpha \geq 1$  then (H) holds, with a bound depending on  $\alpha$  and  $c$  alone.*

*Proof.* (i) The  $a_i$  are distinct odd integers, so  $a_k \geq 2k - 1$ ; with  $a_k \asymp ck^\alpha$  this forces  $c \gg k^{1-\alpha}$ , so  $a_1 \asymp c \rightarrow \infty$ . For  $d < a_1/2$  no  $a_i$  is  $\leq 2d$ , hence  $N_d = 0$ ,  $\sigma_d = 0$ ,  $\delta_d = d$ , and

$$\sum_{d \geq 1} 2^{-N_d} \delta_d^2 \geq \sum_{d < a_1/2} 2d^2 \asymp a_1^3 \asymp k^{3(1-\alpha)} \rightarrow \infty.$$

(ii) For  $\alpha \geq 1$  one has  $a_1 = O(1)$  and  $N_d = \#\{i : a_i \leq 2d\} \gg d^{1/\alpha}$ , while  $\delta_d = d + \sigma_d/2 \ll d^{1+1/\alpha}$ , so the general term is  $\ll 2^{-c'd^{1/\alpha}} d^{2+2/\alpha}$ , which is summable: an exponential in a positive power of  $d$  beats a polynomial. The bound involves  $\alpha$  and  $c$  and not  $k$ .  $\square$

Both halves are visible in the computation. For  $\alpha = 1/2$  the series reads 4923, 10245, 16608, 24464 at  $k = 60, 100, 140, 180$ , growing like  $k^{3/2}$  — which is  $a_1^3 = k^{3(1-\alpha)}$  with  $\alpha = 1/2$ , exactly as (i) predicts. For  $\alpha \geq 1$  it is flat in  $k$  to every digit printed:

| profile                      | odds | $a_i \asymp i^{3/2}$ | squares | cubes              | $\alpha = 1/2$ |
|------------------------------|------|----------------------|---------|--------------------|----------------|
| $\alpha$                     | 1    | $3/2$                | 2       | 3                  | $1/2$          |
| $\sum_d 2^{-N_d} \delta_d^2$ | 69.5 | 266.07               | 6473.4  | $1.112 \cdot 10^7$ | <i>grows</i>   |

[experimentally confirmed,  $k = 60, \dots, 180$ ; [alpha\\_r092](#)] Proposition 6.1 is about clean powers, and the primes do not route through it: Theorem 3.1 for the primes quotes §4.2, and the primes' (H) is checked directly where (H) is stated, from  $N_d = \pi(2d) - 1 \gg d/\log d$ . No reader need look for  $i \log i$  in part (ii). So **(H) cuts the power profiles at exactly  $\alpha = 1$** , and the cubes sit inside the admissible range but beyond the reach of computation.

## 7 Honest scope

In the style of paper 1 §10 and paper 2 §10, here is what rests on what.

- *Formally verified*: the combinatorial layer, inherited from paper 1, for every target.
- *Proved*: the bridge of §4.2 — Lemma 4.8 in full, Lemma 4.10 in all three parts (with (c) quoted verbatim from paper 2's Step 2 at  $n = 1$ , and the covering apparatus of that paper not needed here, see Remark 4.16), Lemma 4.17, and Proposition 6.1. The minor-arc input is  $m$ -independent, so it transfers to off-centre targets unchanged; paper 2's ineffectivity through the Siegel–Walfisz constant is inherited at the single point recorded in Remark 4.13.
- *Proof skeleton, with the analytic ingredients in place*: Theorems 3.1 and 4.1. Regions R2 and R3 of Proposition 4.20 are proved — they quote the bridge — and region R1 is the classical local limit computation with  $p_a$ -weights, given here to the level of its cumulant budget rather than with every Edgeworth constant written out. That is the one gap between this paper and a complete proof, and it is a gap of exposition rather than of ingredients.
- *Experimentally confirmed, with the range stated*: rigidity (§3), the transfer function to 0.8–1.9% of the signal (Table 1), universality (§5), the profile-dependent constant  $4(2\alpha + 1)/(\alpha + 1)$  verified on four profiles.
- *Not claimed*: the residual 0.8–1.9% of Table 1 is consistent with finite size and shrinks in  $k$ , but we do not claim it is zero.

*Remark 7.1* (the single ineffective constant). Precisely: every estimate used here is effective except part (b) of Lemma 4.10, whose error term is the Siegel–Walfisz error, and paper 2’s appendix identifies the same constant  $C_1$  as the only ineffective one in that paper, entering there at one point as well. Paper 1 is unconditional and effective throughout. **Ineffective is not conditional**: no statement here depends on an unproved hypothesis; what is not available is a numerical threshold beyond which the asymptotics take hold.

*Remark 7.2* (approaches that did not work). As in paper 2 we keep the dead ends, with reasons. *The  $\Gamma_x$  family*: the idea that off-centre targets are described by a family of tilted gap series indexed by  $x$ . Rejected by the data — rewriting the observed collapse in the exact tilt parameter makes it worse, not better. *The multiplicative bridge*:  $|\tilde{G}_B| \leq |G_B|^\kappa$ , which would have carried every exponent of paper 2 across at once. Impossible for the reason in Remark 4.9, and the underlying inequality runs the other way. *The residual as a tilt-versus-slope artefact*: the constants of Remarks 2.5 and 4.7 agree to two digits on every profile, but substituting  $s$  for  $\lambda$  makes the residual three times worse — a shared driver, not a cause. *A half-integer exponent* reported in an early draft, which was an artefact of the Gaussian slope (Remark 2.3). *Restating (H) so that the cubes violate it*: the series is bounded for the cubes and unbounded for  $\alpha < 1$ , and the hypothesis follows the computation rather than the narrative.

## Data availability

Every number in this paper is produced by a script that writes a log beside itself, as in papers 1 and 2. The scripts named in the [status] tags — `table1_r086`, `kappa_r088`, `tilt_r088`, `sres_r088`, `arc_r088`, `eta_r090`, `gap_r090`, `saddle_r092`, `alpha_r092`, `t1row_r094`, `p3_r096`, `phig_r096`, `e8_r103` — and their logs are part of the public artefact.

## References

- [1] K. Amauchi, *The gap series of an integer sequence: an arithmetic invariant governing subset-sum landscapes*. Manuscript, 2026.
- [2] K. Amauchi, *Asymptotic flatness of subset-sum landscapes of primes: the sub-peak spectrum and the constant  $\sqrt{3}/2$* . Manuscript, 2026.